Colloids and Surfaces, 68 (1992) 1–8
Elsevier Science Publishers B.V., Amsterdam

Adsorption of dextrans onto hydroxyapatite: presence of maxima

S. Spaltro, K.P. Ananthapadmanabhan, M. Frushour and M. Aronson
Unilever Research U.S., 45 River Road, Edgewater, NJ 07020, USA

(Received 16 September 1991; accepted 19 May 1992)

Abstract

Hydroxyapatite has been widely used as a model substrate for studying biological systems involving mineralization and protein adsorption. In the present study, hydroxyapatite from three commercial sources was used to study the adsorption behavior of commercially available dextrans. Depending on the molecular weight of dextran and the technique used for dispersing the solid, maxima were observed in the adsorption isotherms for all three adsorbents. Scanning electron microscopy observations showed significant differences in the morphological properties of the hydroxyapatite samples. The adsorption maximum is attributed to the inability of the polymer molecules to diffuse into the interior of hydroxyapatite aggregates. The adsorption results have been analyzed to estimate the pore volume and the average pore size in the aggregates.

Keywords: Adsorption maxima; dextran adsorption; hydroxyapatite; polymer adsorption; polymer exclusion.

Introduction

Hydroxyapatite has been widely used as a model susbtrate for studying biological systems involving mineralization, protein adsorption and biofilm formation. The behavior of hydroxyapatite in these systems can be expected to be influenced markedly by the chemical, electrokinetic and morphological properties of the solid surface [1]. These properties can be especially important in adsorption studies involving proteins, macromolecules and surfactant micelles. In polymer adsorption systems, for example, certain molecular weight fractions of the polymer may be excluded from a porous substrate. In competitive adsorption studies involving a polymer and an inorganic species, however, the low molecular weight species will be able to penetrate into the pores and the polymer may be excluded preferentially. Recognition of these effects is extremely important in making a meaningful interpretation of the experimental results. In the present study, hydroxyapatite samples from commercial sources were used to determine the appropriate substrate and the methods needed to understand the adsorption behavior of different molecular weight dextrans from aqueous solutions.

Correspondence to: K.P. Ananthapadmanabhan, Unilever Research U.S., 45 River Road, Edgewater, NJ 07020, USA.

Experimental

Materials

Hydroxyapatite samples were purchased from the following three sources: Biorad (Biogel HTP), Aldrich, and Albright & Wilson (U.K.). Dextran of average molecular weight 15 000–20 000 was obtained from Polyscience Inc. The dextrans of molecular weights 6000 and 500 000 were from Fluka Chemicals.

0166-6622/92/$05.00 © 1992 — Elsevier Science Publishers B.V. All rights reserved.

Adsorption experiments

The following two protocols were employed.

Method I

The amount of polymer adsorbed on hydroxyapatite was determined by the following solution depletion technique. Hydroxyapatite (1 g) was added to 8 ml of polymer solution (in 0.05 *M* sodium chloride solution, pH 7–7.5) to give 16.0–160.0 mg of dissolved polymer per gram of hydroxyapatite. The sample was placed in an end-to-end rotator and incubated at 37°C for 90 min. The suspension was centrifuged and the supernatant was analyzed for remaining polymer using the orcinol method [2]. The amount adsorbed was computed as the difference between the initial and final concentrations of the polymer.

Method II

Hydroxyapatite was suspended in 0.05 *M* sodium chloride solution and stirred vigorously with a magnetic stirring bar for 16 h. The slurries were incubated with a polymer solution in an end-to-end rotator to give a concentration ranging from 20.0 to 200.0 mg of polymer per gram adsorbent for 90 min at 37°C. The amount adsorbed was determined as in Method I.

Surface area

The surface area of each adsorbent was determined on a Flowsorb II 2300 instrument (Micrometrics) using a single-point Brunauer–Emmett–Teller (BET) technique. The surface area obtained was an average value of at least three adsorption–desorption experiments. The surface area for each sample is as follows: Biogel HTP, 71 $m^2 g^{-1}$; Aldrich, 79 $m^2 g^{-1}$; Albright & Wilson, 29 $m^2 g^{-1}$.

Scanning electron microscopy (SEM)

Information on the surface morphology of the hydroxyapatite samples was obtained using the JEOL T-300 scanning electron microscope at various magnifications and operating voltages. Since the samples were non-conductive, they were coated with layers of gold prior to the SEM measurements.

Mercury porosimetry

The average pore size and pore volume of Biogel HTP were measured by mercury-intrusion porosimetry which can detect porosity in the range 4 nm to 100 μm. The surface tension and contact angle values used for mercury were 480 erg cm^{-2} and 140° respectively. The total pore volume intruded was 0.6538 ml g^{-1} at 33 000 psi; pores were noted at about 8 μm and 200 Å. The average pore diameter was determined to be around 200 Å.

Transmission electron microscopy (TEM)

Samples were prepared in the following manner. A slurry of the powder in ethanol was ground and ultrasonicated until the powder was sufficiently fine. This was necessary to deaggregate the big chunks so that the fine structure can be imaged clearly. A carbon-coated TEM grid was then dipped into the diluted slurry to deposit the hydroxyapatite powder on the grid surface. The surface was examined in a JEOL 100C transmission electron microscope using an accelerating voltage of 100 kV. The texture or porosity present in the hydroxyapatite powder was imaged by the "defocus-contrast" technique. Defocus-contrast micrographs show features which exhibit reversal of contrast, indicating that the features are indeed phase objects such as voids. The contrast arises from the difference in the mean inner potential between the pores and matrix material. By focusing either above or below the sample, a phase difference can be induced between the electron beams. This phase difference manifests itself as either dark or light rings which surround the void. The voids in the Biogel HTP were found to possess a lamellar morphology and their sizes range from 150 to 1000 Å.

Results and discussion

The adsorption isotherms of different molecular weight dextrans on Biogel HTP are shown in Fig. 1. It can be seen that for dextran-6K, i.e. the dextran of molecular weight 6000, adsorption increases with increasing concentration and attains a plateau value above an initial polymer concentration of about 12 mg ml^{-1} hydroxyapatite. Interestingly, both the dextran-15K and dextran-500K exhibit an adsorption maximum in their isotherms. Furthermore, these results clearly show that adsorption is lower for the higher molecular weight polymers. The presence of a maximum as well as lower adsorption at higher molecular weights is contrary to what is expected for polymer adsorption on non-porous solids. However, exclusion of higher molecular weight polymers from micropores can indeed lead to lower adsorption.

The presence of micropores was verified by examining the morphology of the particles using SEM (Fig. 2). It is evident that the Biogel HTP sample exhibits a "cake-like" structure with an average particle size of at least 100 μm. Attempts to break the aggregates by vigorous stirring and even sonication were not successful in the sense that these sonicated particles did not eliminate the adsorption maximum for dextran-500K.

Adsorption results obtained under mild agitation conditions (see Experimental section, Method 1), using two other hydroxyapatite samples, are shown in Figs 3 and 4 and also exhibit a maximum for dextran-500K. The maximum adsorption density values for these samples are much higher than those obtained using Biogel HTP. SEM pictures of these samples show that these, unlike the Biogel HTP, are powder-like aggregates (Figs 5 and 6). Interestingly, a thorough dispersion of these particles, obtained by stirring for 16 h, resulted in an isotherm with almost no maximum (Fig. 7).

It is clear from the results obtained that the presence of a maximum is directly related to the cake-like structure of Biogel HTP. Furthermore.

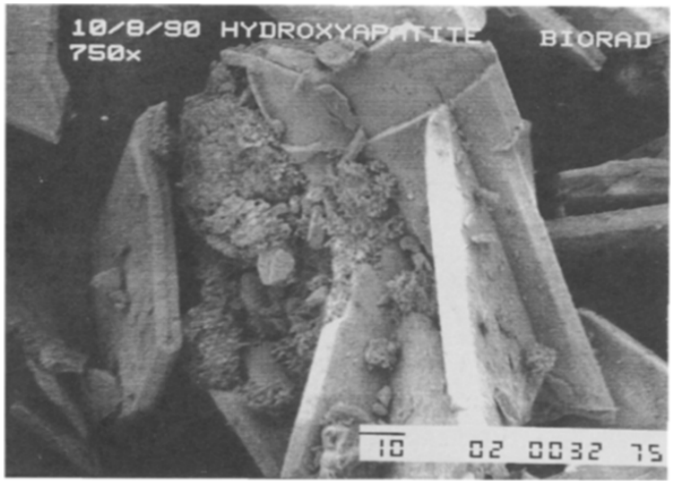


Fig. 2. Electron micrographs of Biogel HTP.

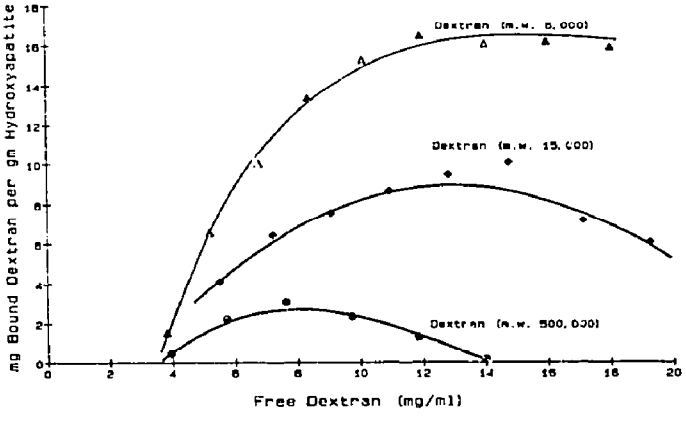


Fig. 1. Adsorption of dextrans on Biogel HTP (Biorad).

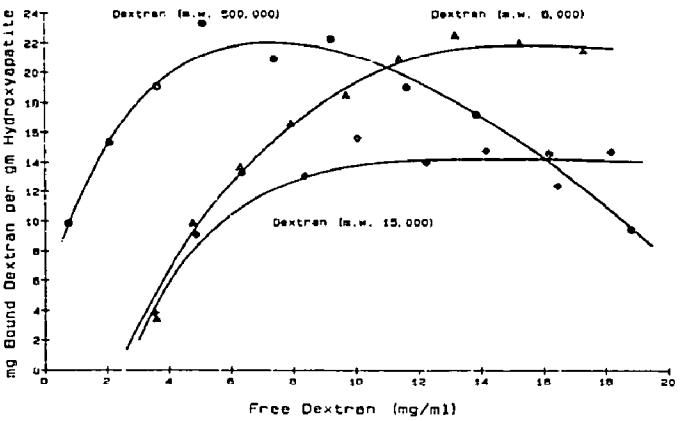


Fig. 3. Adsorption of dextrans on the hydroxyapatite from Aldrich.

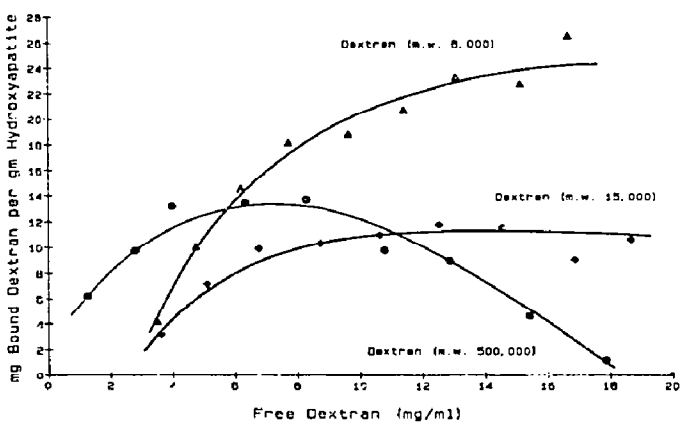


Fig. 4. Adsorption of dextrans on the hydroxyapatite from Albright & Wilson.

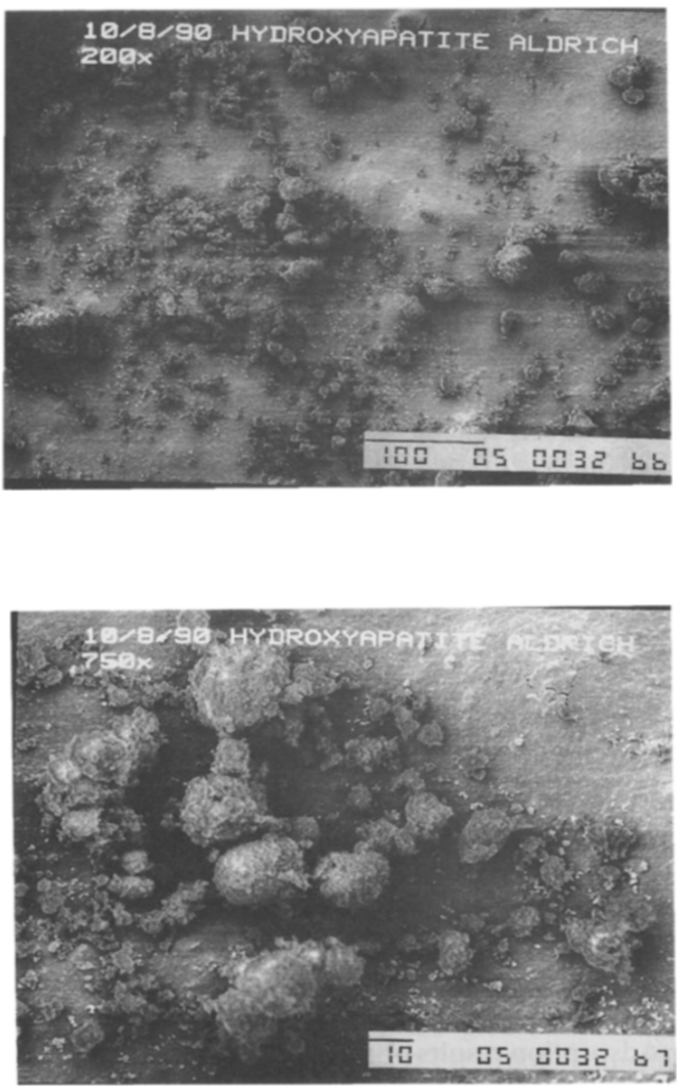


Fig. 5. Electron micrographs of the hydroxyapatite from Aldrich.

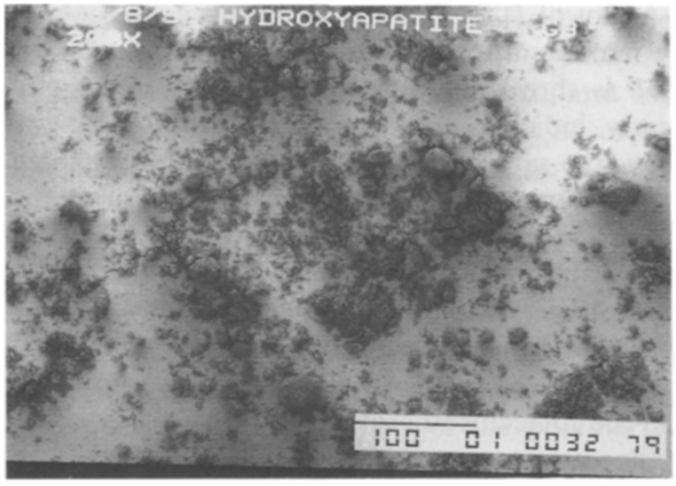
even powder-like samples, if improperly dispersed, can lead to anomalous results.

Intuitively, one would expect exclusion of polymers from pores to result in a lower plateau value than in the case where the pores are large enough to allow the polymers to penetrate so that all the internal area of the aggregate is available for adsorption. The reason for the presence of a maximum in the isotherm is not obvious from the fact that the substrate is porous. The phenomenon can be explained by examining the definition of thermodynamic adsorption in terms of what is actually measured by the solution depletion technique and how much is really on the surface. In classical colloid chemistry, adsorption refers to the Gibbs thermodynamic adsorption which is defined as the surface excess, i.e. the excess solute in a unit volume at the interfacial region over that in the same volume in the bulk solution. This can be expressed as

$$\text{Adsorption} = C_s - C_b \tag{1}$$

where C_s refers to the solute concentration at the surface and C_b is the bulk concentration.

Note that according to the above definition, if for some reason the surface repels or excludes a solute from the interfacial region, a C_s which is lower than C_b, can result and consequently the



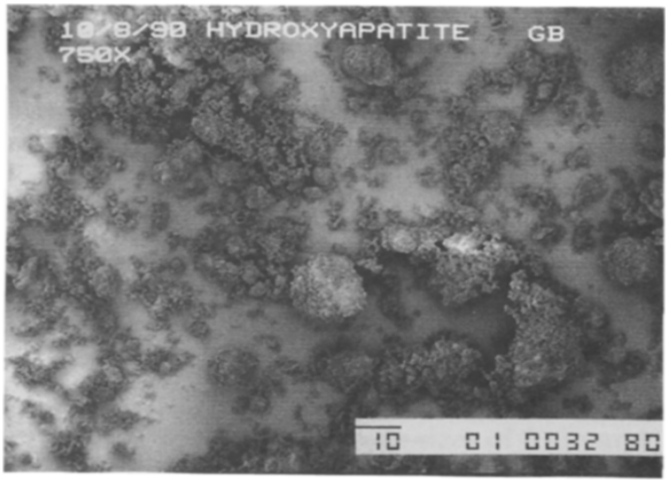


Fig. 6. Electron micrographs of the hydroxyapatite from Albright & Wilson.

adsorption can be negative. The negative adsorption will continue to increase with increasing bulk concentration. Factors which can cause solute exclusion include electrostatic repulsion of similarly charged adsorbate and substrate, size exclusion as well as solvation/desolvation of the adsorbate and the adsorbent [3]. In the case of size exclusion, adsorption can occur only on the external surface of a porous substrate in which case C_s will initially increase and ultimately reach a plateau value. The adsorption value will correspondingly increase due to C_s up to the plateau

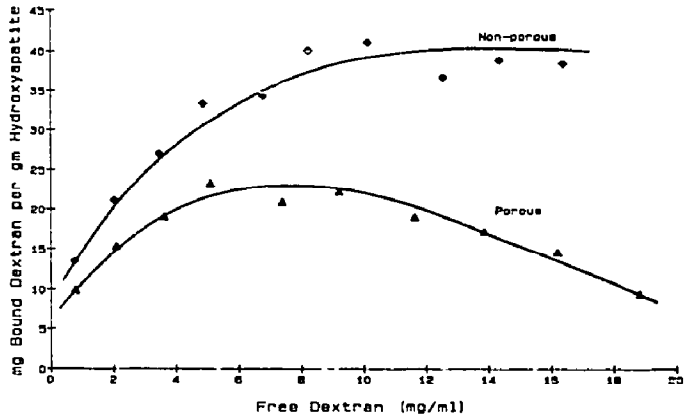


Fig. 7. Comparison of the adsorption isotherm for dextran-500K on porous and non-porous adsorbent.

concentration but above this concentration the increase in C_b will result in a decrease in adsorption. This is the phenomenon which gives rise to an adsorption maximum in this case.

The determination of adsorption using the solution depletion technique involves taking the difference between the concentrations of the solute before and after contact with the solid. This difference is then attributed to surface adsorption. It is then normalized on a "per unit area" or "per gram" basis. Adsorption measured using this technique is, therefore, the thermodynamic or Gibbs adsorption. If, however, the adsorbed amount is measured by directly analyzing the solid, the result obtained will correspond to the actual physical accumulation on the solid. The latter is referred to as the Guggenheim adsorption. Normally, the distinction between the two types of adsorption is not made, simply because for most surface-active substances the surface concentration is markedly higher than the bulk concentration and, therefore, the C_b term is negligible compared to C_s.

An ideal situation in which polymer molecules adsorb on the external surface, and are excluded completely from pores, can be modeled as follows. The adsorption density, Γ, on a per gram basis is given by

$$\Gamma = (C_i - C_f)/S \tag{2}$$

where C_i is the initial concentration and C_f is the

final concentration, both in milligrams of solute per millilitre of solution, Γ is the number of milligrams of polymer adsorbed per gram of adsorbent and S is the adsorbent concentration in grams per millilitre of solution.

If the adsorbent is porous such that all the solute is excluded and the solvent is not, then the actual initial concentration will be higher than C_i and is given by

$$C_i V_i = C_i'(V_i - V_p S V_i) \qquad C_i = C_i'(1 - V_p S) \tag{3}$$

where V_i is the initial volume, C_i' is the corrected polymer concentration in solution and V_p is the average pore volume in millilitres per gram of the solid.

The calculated apparent adsorption, Γ_a, using the uncorrected initial concentrations is

$$\Gamma_a = (C_i - C_f)/S \tag{4}$$

Substituting for C_i in terms of C_i',

$$\Gamma_a = [C_i'(1 - V_p S) - C_f]/S$$
$$= [(C_i' - C_f)/S] - C_i' V_p \tag{5}$$

The first term is the true Gibbs adsorption density Γ and the second term is the porosity effect

$$\Gamma_a = \Gamma - C_i' V_p \tag{6}$$

Substituting for C_i' in terms of the known C_i and rearranging, one can write for concentrations above the saturation-adsorption density, Γ_s,

$$\Gamma_a = \Gamma_s - K C_i \tag{7}$$

where

$$K = V_p/(1 - V_p S) \tag{8}$$

Thus a plot of Γ_a vs C_i under saturation-adsorption conditions will result in a straight line with the gradient, K, giving information on the porosity of the solid. K depends on the properties of the solid and its concentration in solution. It is clear from Eqn (7) that the apparent adsorption at concentrations above the saturation concentration will decrease linearly with increasing bulk concentration.

A plot of Γ_a vs C_i for concentrations well above the maximum for dextran-500K on porous Biogel HTP is shown in Fig. 8. The linearity of the plot clearly indicates that the levels chosen are well above the saturation concentration. The estimated pore volume is 1.4 ml g^{-1} and the value of the saturation adsorption is 32.2 mg g^{-1} (Fig. 8). Using these values and Eqns (2) and (3), the true Gibbs adsorption is estimated for dextran-500K and is shown in Fig. 9. It can be seen when comparing the adsorption isotherm of dextran-500K in Fig. 1 and the correct isotherm in Fig. 9 that the presence of porous structures in an adsorbent can give misleading results. The estimated true Gibbs adsorption is more consistent with the general observation that higher molecular weight polymers

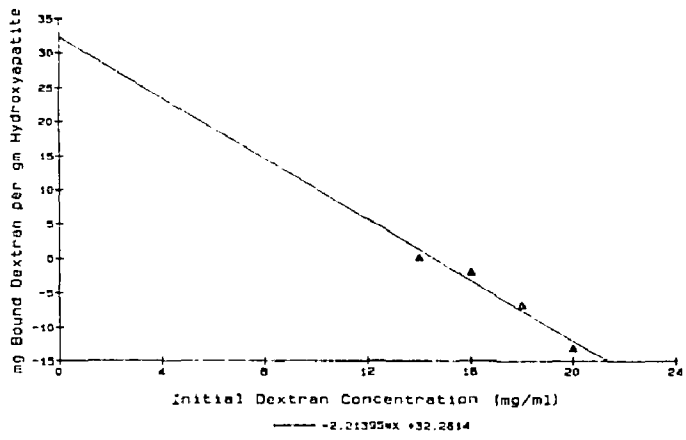


Fig. 8. Plot of Γ_a vs concentration from the adsorption isotherm of dextran-500K on Biogel HTP.

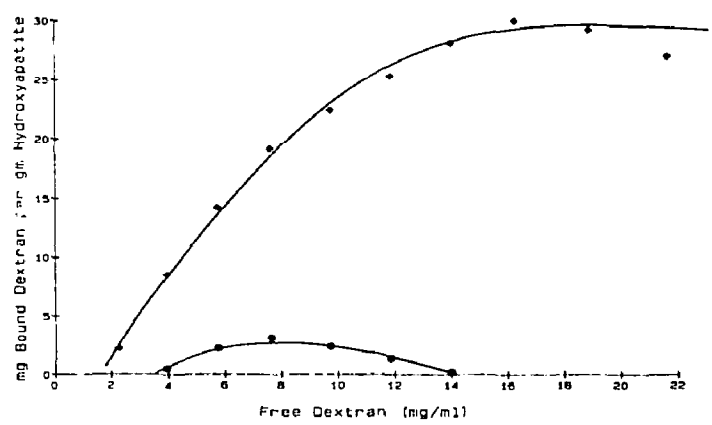


Fig. 9. Corrected Gibbs adsorption of dextran-500K on Biogel HTP. (◆) Corrected adsorption; (●) measured adsorption.

adsorb more than their lower molecular weight counterparts.

Assuming the pores to be cylindrical, and using the estimated pore volume from the adsorption isotherm and the measured surface area by the nitrogen adsorption technique, the pore diameter of the Biogel HTP is estimated to be 800 Å. We have attempted to compare the calculated pore diameter with the radius of gyration of the polymer in solution. Assuming the polymer to be linear, for dextran-500K, the radius of gyration (R_g) and the hydrodynamic diameter (D_1) of the polymer can be calculated using the following equations [4,5]

$$R_g = 0.23\ M^{0.50} \tag{9}$$

$$D_1 = 2\sqrt{6}R_g \tag{10}$$

R_g is found to be 162.6 Å and D_1 is 797.0 Å.

It is clear that these dimensions are indeed comparable to the estimated pore diameter and, therefore, the polymer molecule can be expected to be excluded from such pores. These results are in agreement with the TEM results which show that the pore-size range is between 150 and 1000 Å (Table 1).

We have also attempted to determine the pore size and pore-size distribution using mercury porosimetry and TEM. The former indicated two groups of pores with average sizes of 8 μm and 200 Å. The 8 μm pores may well be a result of the void in the aggregates as seen in SEM. This pore-size range is too big to exclude any of the polymers in this study and would not play any role in influencing the adsorption behavior of the polymers. TEM gave pore sizes ranging from 150 to 1000 Å. It is most reasonable that dextran-500K with a calculated diameter of about 800 Å would be totally excluded from all the micropores. The dextran-15–20K with a calculated diameter of 138–159 Å would be partially excluded, especially from the pores of 200 Å, resulting in an adsorption maximum (Fig. 1). Dextran-6K with a calculated diameter of ≈90 Å could gain access to all the pores and this is in agreement with the observation that the corresponding isotherm does not exhibit a maximum.

It is interesting that the results obtained from mercury porosimetry and TEM are indeed in accordance with the pore-size estimates from the adsorption data and the dimensions of the polymer molecule in solution. Note, however, that the pore volume estimated from mercury porosimetry data is about 40% of the value obtained from polymer adsorption results. The reason for such a big difference is not clear at present. Partial breakdown of some of the internal structure under high pressure during mercury porosimetry can possibly account for some of the observed difference.

TABLE 1

Characterization of Biogel HTP

Technique	Pore size (Å)	Surface area ($m^2\ g^{-1}$)	Pore volume ($ml\ g^{-1}$)
Mercury porosimetry	200[a]	65.4	0.65
TEM	150–1000		
Adsorption isotherm			
Dextran-500K	<888		1.4
Dextran-15–20K	≈150		
N_2 adsorption		71	

[a]Average value.

Conclusions

Under gentle mixing of the solid with a polymer solution, adsorption isotherms as a function of polymer concentrations exhibited a maximum for each hydroxyapatite sample for dextran-500K. The presence of maxima is indicative of some form of polymer exclusion and preferential adsorption of the solvent. SEM of two hydroxyapatite samples indicated a powder-like structure whereas the third sample showed a cake-like morphology. The first two samples could be broken down by intense mixing, as seen by photomicrographs, and produced a normal adsorption isotherm after this treatment. The cake-like sample did not break down even after sonication as judged by the amount of polymer adsorbed before and after

sonication. The calculated pore diameter from the adsorption isotherm of an excluded polymer agreed well with the experimentally determined results using mercury porosimetry and TEM. It is clear that the adsorption maximum is due to the exclusion of the high molecular weight polymer from the micropores. It is evident from this study that thorough investigation of the morphological properties of the adsorbent is necessary in generating meaningful and reproducible adsorption data.

Acknowledgment

The authors acknowledge the contribution of Dr. W. Bahary for the determination of molecular size, Ms. K. Hoyberg for the SEM Work and Dr. H. Kerch for the TEM and mercury porosimetry work. The authors also thank Unilever Research U.S. for permission to publish the work.

References

1 D.N. Misra (Ed.), Adsorption on and Surface Chemistry of Hydroxyapatite, Plenum Press, New York, 1984.
2 R.J. Winzler, in D. Glick (Ed.), Methods in Biochemical Analysis, Vol.2, Interscience, London, 1955, p. 290.
3 K.P. Ananthapadmanabhan and P. Somasundaran, Colloids Surfaces, 7 (1983) 105.
4 W.W. Yau, J.J. Kirkland and D.D. Bly (Eds), Modern Size-Exclusion Liquid Chromatography, Wiley, New York, 1979, p. 31.
5 J. Brandrup and E.H. Immergut (Eds), Polymer Handbook, Wiley, New York, 1989, p. VII-32.